IPL

April 2, 2026

[1]:
```
import numpy as np

#Seasons
Seasons =␣
↪["2010","2011","2012","2013","2014","2015","2016","2017","2018","2019"]
Sdict = {"2010":0,"2011":1,"2012":2,"2013":3,"2014":4,"2015":5,"2016":6,"2017":
↪7,"2018":8,"2019":9}

#Players
Players =␣
↪["Sachin","Rahul","Smith","Sami","Pollard","Morris","Samson","Dhoni","Kohli","Sky"]
Pdict = {"Sachin":0,"Rahul":1,"Smith":2,"Sami":3,"Pollard":4,"Morris":
↪5,"Samson":6,"Dhoni":7,"Kohli":8,"Sky":9}

#Salaries
Sachin_Salary =␣
↪[15946875,17718750,19490625,21262500,23034375,24806250,25244493,27849149,30453805,23500000]
Rahul_Salary =␣
↪[12000000,12744189,13488377,14232567,14976754,16324500,18038573,19752645,21466718,23180790]
Smith_Salary =␣
↪[4621800,5828090,13041250,14410581,15779912,14500000,16022500,17545000,19067500,20644400]
Sami_Salary =␣
↪[3713640,4694041,13041250,14410581,15779912,17149243,18518574,19450000,22407474,22458000]
Pollard_Salary =␣
↪[4493160,4806720,6061274,13758000,15202590,16647180,18091770,19536360,20513178,21436271]
Morris_Salary =␣
↪[3348000,4235220,12455000,14410581,15779912,14500000,16022500,17545000,19067500,20644400]
Samson_Salary =␣
↪[3144240,3380160,3615960,4574189,13520500,14940153,16359805,17779458,18668431,20068563]
Dhoni_Salary =␣
↪[0,0,4171200,4484040,4796880,6053663,15506632,16669630,17832627,18995624]
Kohli_Salary =␣
↪[0,0,0,4822800,5184480,5546160,6993708,16402500,17632688,18862875]
Sky_Salary =␣
↪[3031920,3841443,13041250,14410581,15779912,14200000,15691000,17182000,18673000,15000000]
#Matrix
```

```
Salary = np.array([Sachin_Salary, Rahul_Salary, Smith_Salary, Sami_Salary,␣
 ↪Pollard_Salary, Morris_Salary, Samson_Salary, Dhoni_Salary, Kohli_Salary,␣
 ↪Sky_Salary])

#Games
Sachin_G = [80,77,82,82,73,82,58,78,6,35]
Rahul_G = [82,57,82,79,76,72,60,72,79,80]
Smith_G = [79,78,75,81,76,79,62,76,77,69]
Sami_G = [80,65,77,66,69,77,55,67,77,40]
Pollard_G = [82,82,82,79,82,78,54,76,71,41]
Morris_G = [70,69,67,77,70,77,57,74,79,44]
Samson_G = [78,64,80,78,45,80,60,70,62,82]
Dhoni_G = [35,35,80,74,82,78,66,81,81,27]
Kohli_G = [40,40,40,81,78,81,39,0,10,51]
Sky_G = [75,51,51,79,77,76,49,69,54,62]
#Matrix
Games = np.array([Sachin_G, Rahul_G, Smith_G, Sami_G, Pollard_G, Morris_G,␣
 ↪Samson_G, Dhoni_G, Kohli_G, Sky_G])

#Points
Sachin_PTS = [2832,2430,2323,2201,1970,2078,1616,2133,83,782]
 Rahul_PTS = [1653,1426,1779,1688,1619,1312,1129,1170,1245,1154]
Smith_PTS = [2478,2132,2250,2304,2258,2111,1683,2036,2089,1743]
Sami_PTS = [2122,1881,1978,1504,1943,1970,1245,1920,2112,966]
Pollard_PTS = [1292,1443,1695,1624,1503,1784,1113,1296,1297,646]
Morris_PTS = [1572,1561,1496,1746,1678,1438,1025,1232,1281,928]
Samson_PTS = [1258,1104,1684,1781,841,1268,1189,1186,1185,1564]
Dhoni_PTS = [903,903,1624,1871,2472,2161,1850,2280,2593,686]
Kohli_PTS = [597,597,597,1361,1619,2026,852,0,159,904]
Sky_PTS = [2040,1397,1254,2386,2045,1941,1082,1463,1028,1331]
#Matrix
Points = np.array([Sachin_PTS, Rahul_PTS, Smith_PTS, Sami_PTS, Pollard_PTS,␣
 ↪Morris_PTS, Samson_PTS, Dhoni_PTS, Kohli_PTS, Sky_PTS])
```

[2]:
```
#The above data is historical dataset
Games
```

[2]:
```
array([[80, 77, 82, 82, 73, 82, 58, 78,  6, 35],
       [82, 57, 82, 79, 76, 72, 60, 72, 79, 80],
       [79, 78, 75, 81, 76, 79, 62, 76, 77, 69],
       [80, 65, 77, 66, 69, 77, 55, 67, 77, 40],
       [82, 82, 82, 79, 82, 78, 54, 76, 71, 41],
       [70, 69, 67, 77, 70, 77, 57, 74, 79, 44],
       [78, 64, 80, 78, 45, 80, 60, 70, 62, 82],
       [35, 35, 80, 74, 82, 78, 66, 81, 81, 27],
       [40, 40, 40, 81, 78, 81, 39,  0, 10, 51],
```

```
        [75, 51, 51, 79, 77, 76, 49, 69, 54, 62]])
```

[3]:
```
Pdict
```

[3]:
```
{'Sachin': 0,
 'Rahul': 1,
 'Smith': 2,
 'Sami': 3,
 'Pollard': 4,
 'Morris': 5,
 'Samson': 6,
 'Dhoni': 7,
 'Kohli': 8,
 'Sky': 9}
```

[4]:
```
Sdict
```

[4]:
```
{'2010': 0,
 '2011': 1,
 '2012': 2,
 '2013': 3,
 '2014': 4,
 '2015': 5,
 '2016': 6,
 '2017': 7,
 '2018': 8,
 '2019': 9}
```

[6]:
```
Salary//Games
```

```
/tmp/ipykernel_193/1634212085.py:1: RuntimeWarning: divide by zero encountered
in floor_divide
  Salary//Games
```

[6]:
```
array([[ 199335,  230113,  237690,  259298,  315539,  302515,  435249,
         357040, 5075634,  671428],
       [ 146341,  223582,  164492,  180159,  197062,  226729,  300642,
         274342,  271730,  289759],
       [  58503,   74719,  173883,  177908,  207630,  183544,  258427,
         230855,  247629,  299194],
       [  46420,   72216,  169366,  218342,  228694,  222717,  336701,
         290298,  291006,  561450],
       [  54794,   58618,   73917,  174151,  185397,  213425,  335032,
         257057,  288918,  522835],
       [  47828,   61380,  185895,  187150,  225427,  188311,  281096,
         237094,  241360,  469190],
       [  40310,   52815,   45199,   58643,  300455,  186751,  272663,
```

```
          253992,  301103,  244738],
        [      0,       0,   52140,   60595,   58498,   77611,  234948,
          205797,  220155,  703541],
        [      0,       0,       0,   59540,   66467,   68471,  179325,
                0, 1763268,  369860],
        [  40425,   75322,  255710,  182412,  204933,  186842,  320224,
          249014,  345796,  241935]])
```

[7]:
```
np.round(Salary//Games)
```

```
/tmp/ipykernel_193/3663165759.py:1: RuntimeWarning: divide by zero encountered
in floor_divide
  np.round(Salary//Games)
```

[7]:
```
array([[ 199335,  230113,  237690,  259298,  315539,  302515,  435249,
          357040, 5075634,  671428],
       [ 146341,  223582,  164492,  180159,  197062,  226729,  300642,
          274342,  271730,  289759],
       [  58503,   74719,  173883,  177908,  207630,  183544,  258427,
          230855,  247629,  299194],
       [  46420,   72216,  169366,  218342,  228694,  222717,  336701,
          290298,  291006,  561450],
       [  54794,   58618,   73917,  174151,  185397,  213425,  335032,
          257057,  288918,  522835],
       [  47828,   61380,  185895,  187150,  225427,  188311,  281096,
          237094,  241360,  469190],
       [  40310,   52815,   45199,   58643,  300455,  186751,  272663,
          253992,  301103,  244738],
       [      0,       0,   52140,   60595,   58498,   77611,  234948,
          205797,  220155,  703541],
       [      0,       0,       0,   59540,   66467,   68471,  179325,
                0, 1763268,  369860],
       [  40425,   75322,  255710,  182412,  204933,  186842,  320224,
          249014,  345796,  241935]])
```

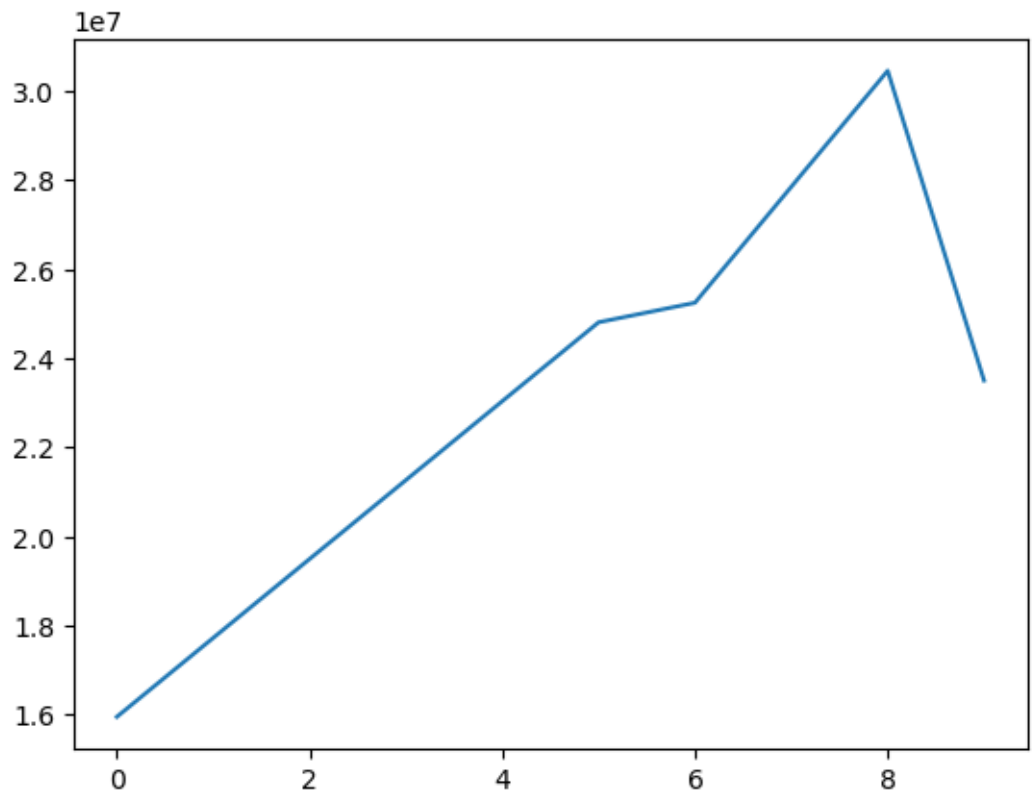
[9]:
```
import matplotlib.pyplot as plt
```

[10]:
```
Salary[0] #will print fist row of 1st column
```

[10]:
```
array([15946875, 17718750, 19490625, 21262500, 23034375, 24806250,
       25244493, 27849149, 30453805, 23500000])
```

[11]:
```
plt.plot(Salary[0]) #1st player salary increase till 2018 & 2019 it is decrease
```

[11]:
```
[<matplotlib.lines.Line2D at 0x7c4d6819dd90>]
```



[12]:
```
Games[0]
```

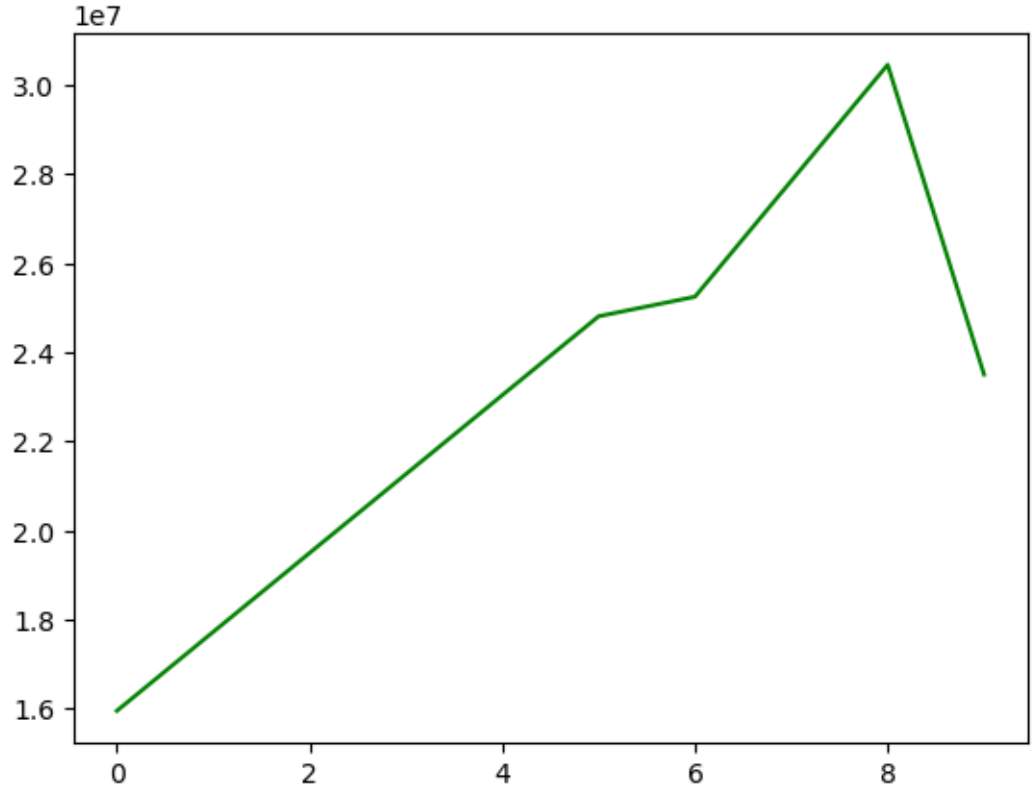
[12]: array([80, 77, 82, 82, 73, 82, 58, 78, 6, 35])

[13]:
```
#Why he play 6 number of match ?
```

[ ]:
```

```

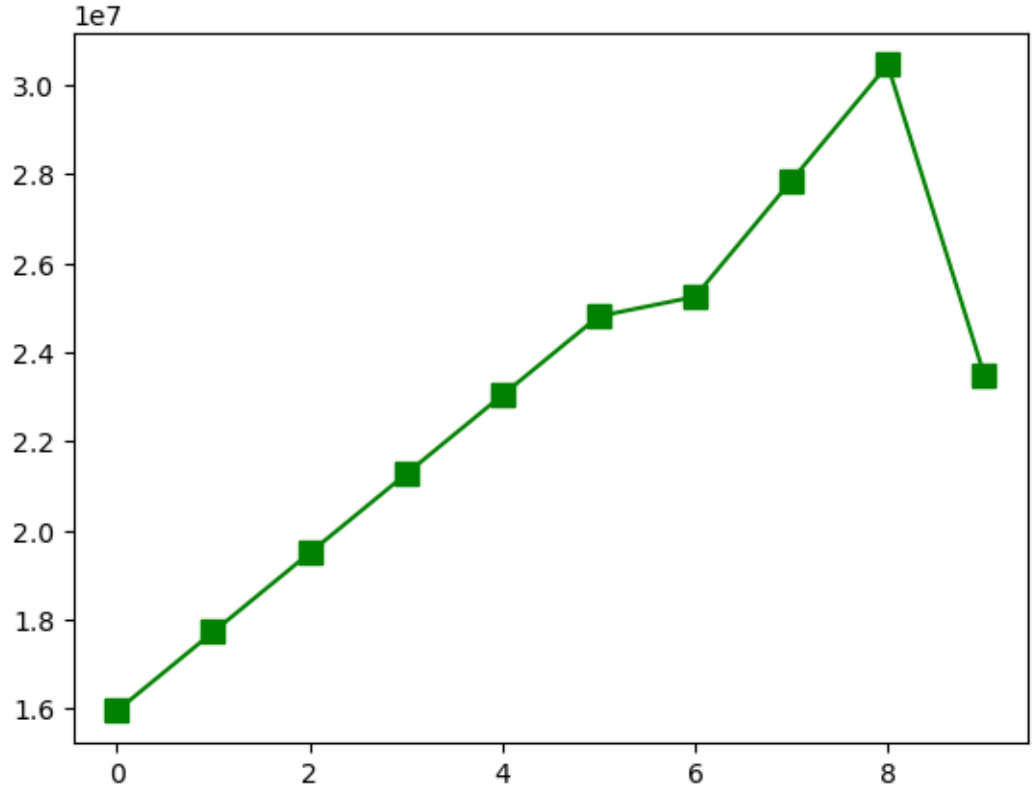
[14]:
```
plt.plot(Salary[0],color='green')
```

[14]: [<matplotlib.lines.Line2D at 0x7c4d68249d30>]



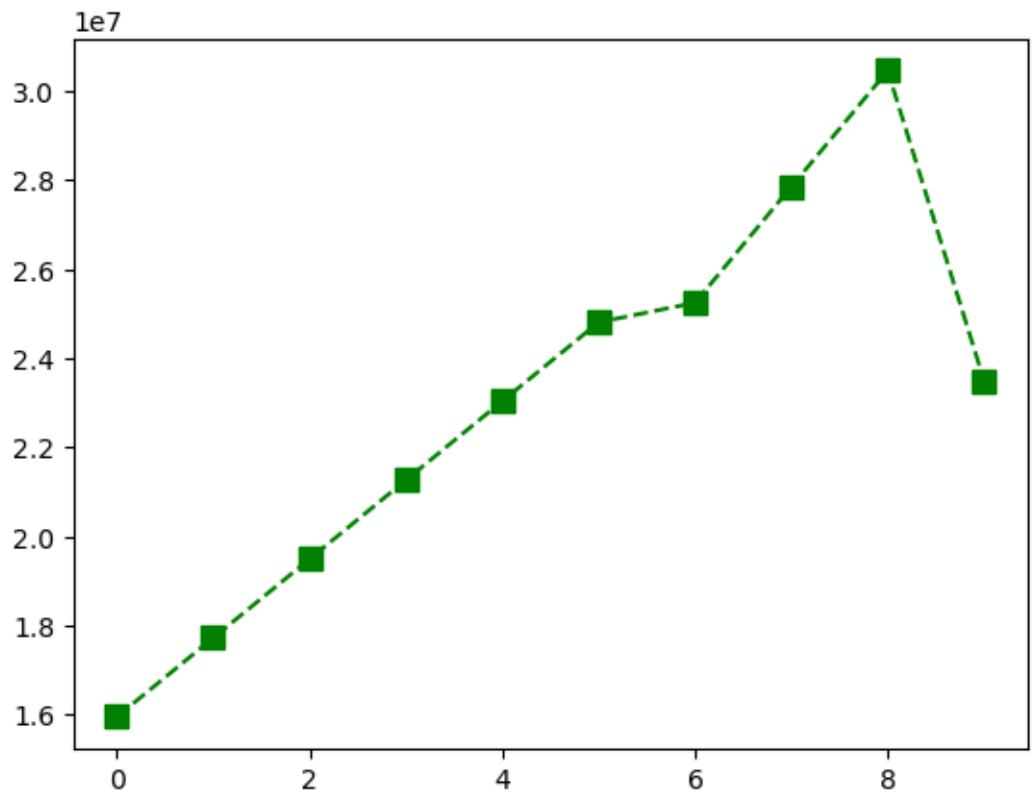
[17]:
```
plt.plot(Salary[0],color='green',marker='s',ms=8) #square box
```

[17]:
```
[<matplotlib.lines.Line2D at 0x7c4d67759220>]
```

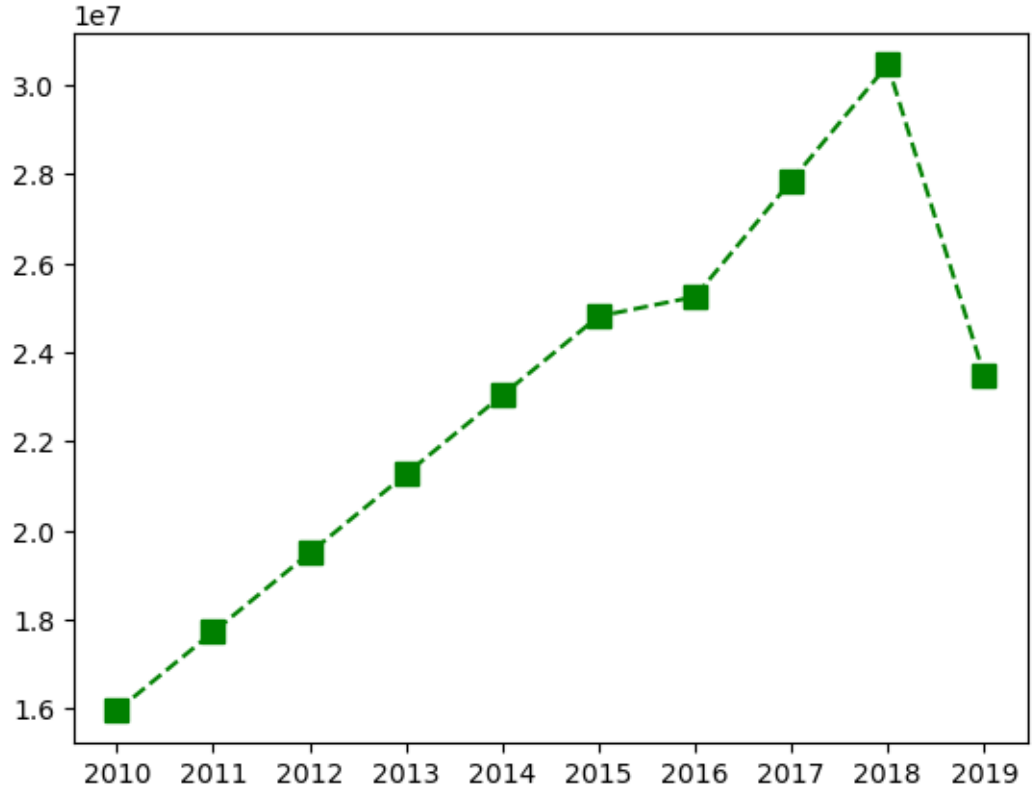


[18]:
```
plt.plot(Salary[0],color='green',ls='--',marker='s',ms=8) #square box
```

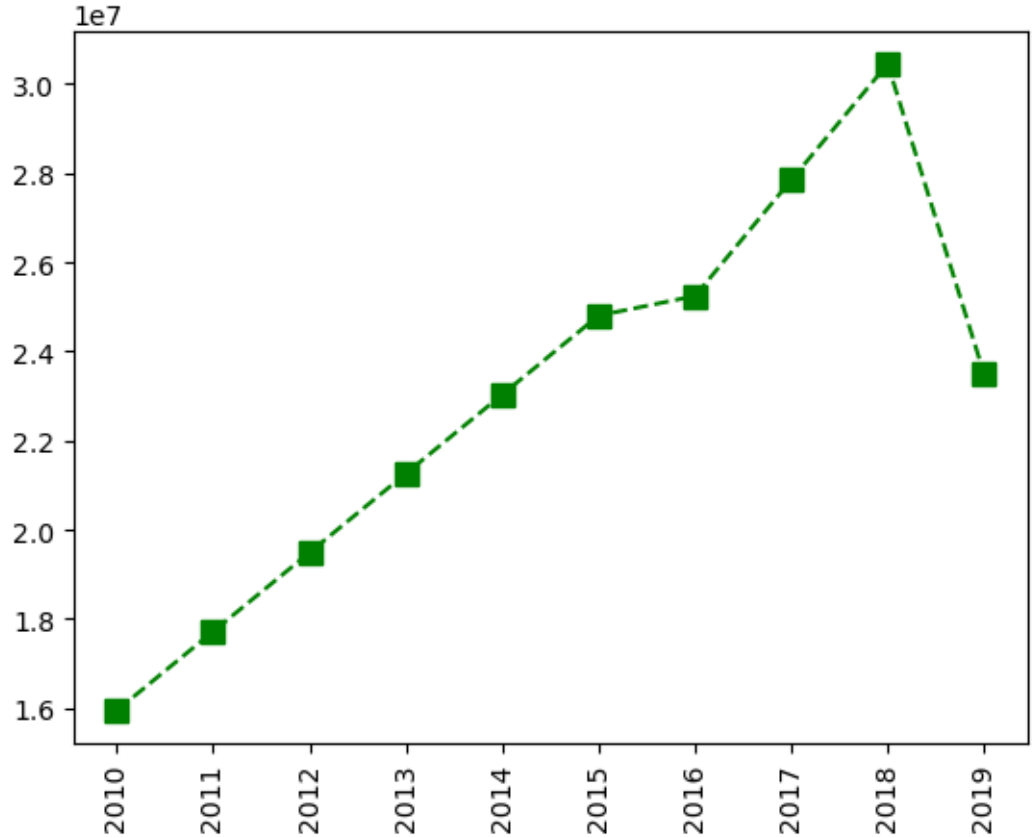
[18]:
```
[<matplotlib.lines.Line2D at 0x7c4d677b3320>]
```



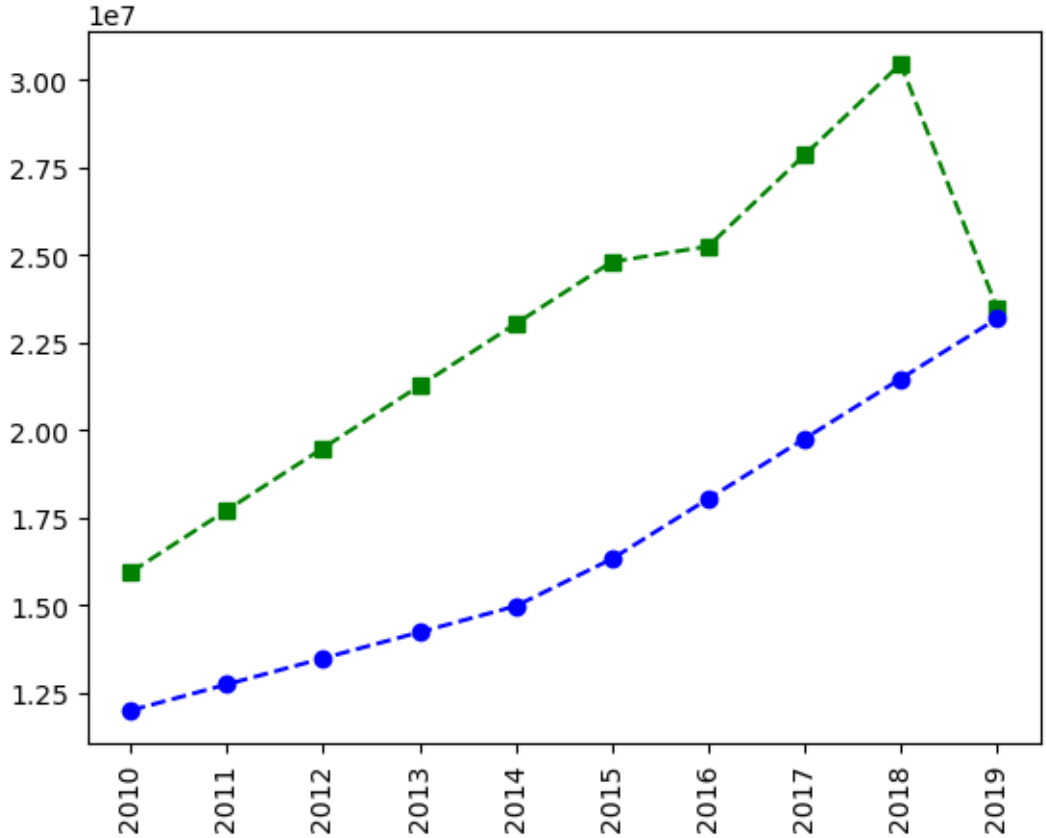
[19]:
```
plt.plot(Salary[0],color='green',ls='--',marker='s',ms=8) #square box
plt.xticks(list(range(0,10)),Seasons) #X axis values
plt.show()
```



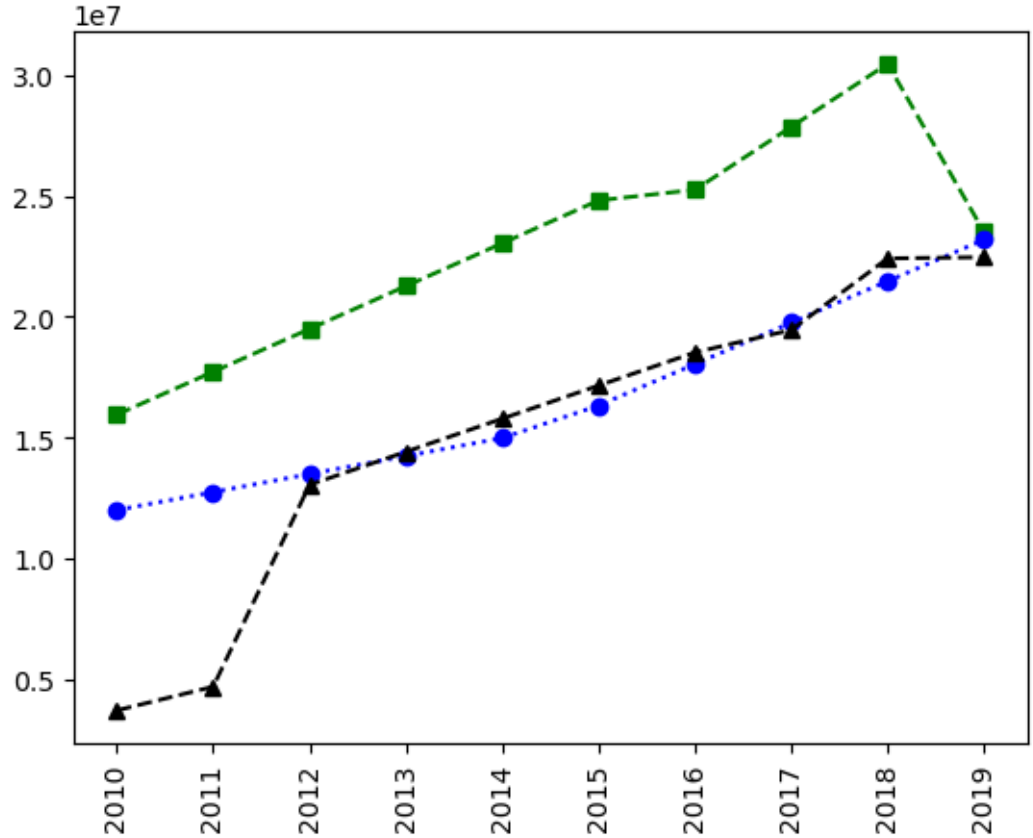
[20]:
```
plt.plot(Salary[0],color='green',ls='--',marker='s',ms=8) #square box
plt.xticks(list(range(0,10)),Seasons,rotation='vertical') #X axis values
#rotation values as 'vertical'
plt.show()
```



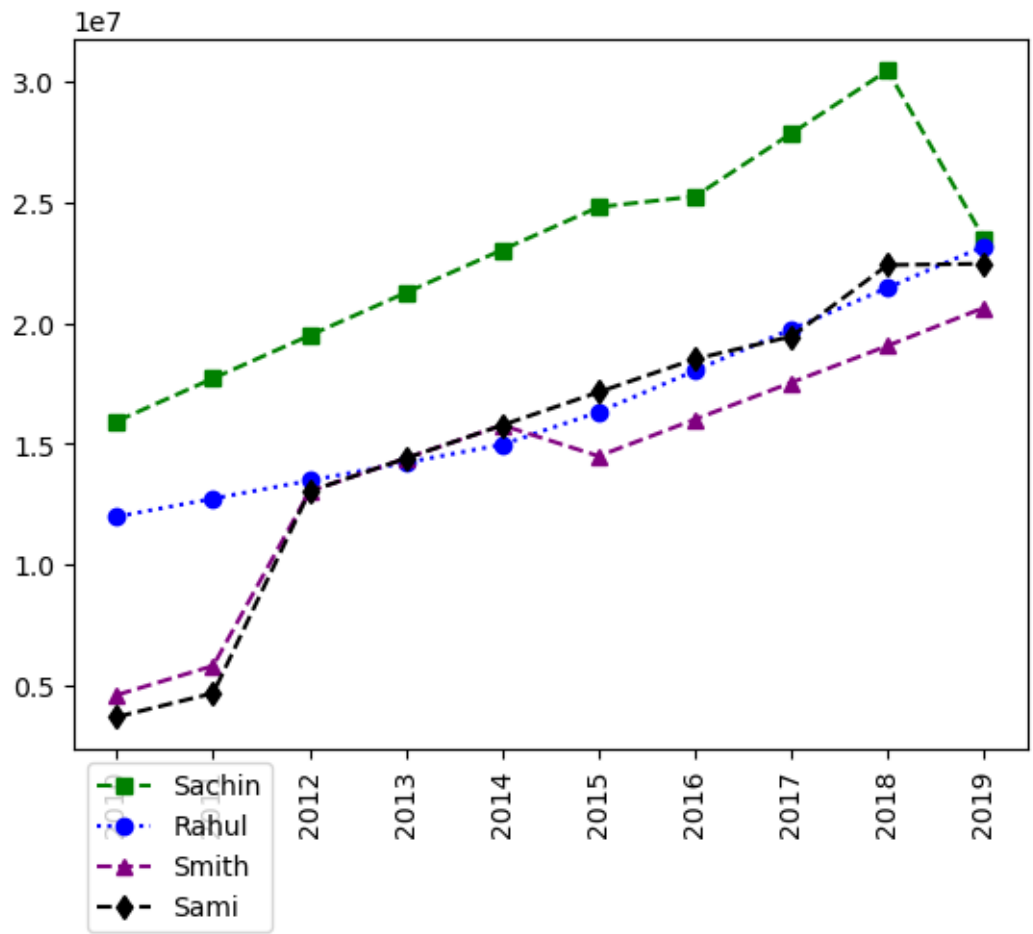
[23]:
```
plt.plot(Salary[0],color='green',ls='--',marker='s',ms=6) #square box
plt.plot(Salary[1],color='blue',ls='--',marker='o',ms=6) #square box
plt.xticks(list(range(0,10)),Seasons,rotation='vertical') #X axis values
#rotation values as 'vertical'
plt.show()
```



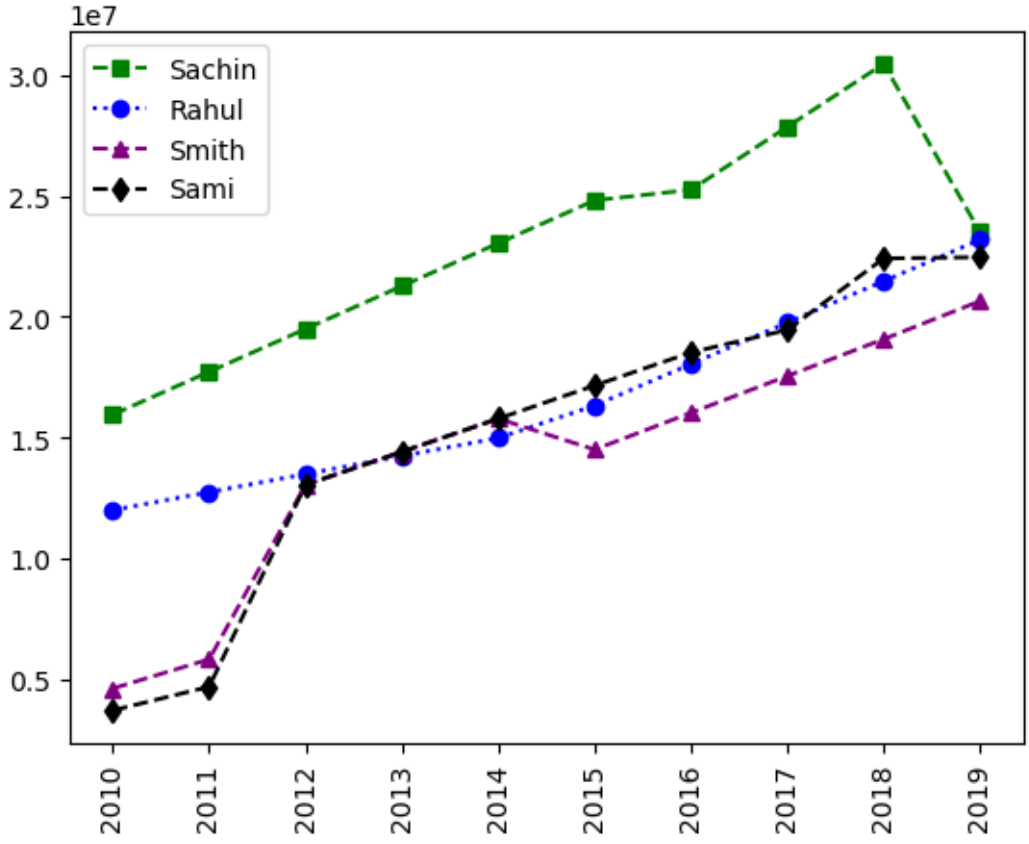
[24]:
```
plt.plot(Salary[0],color='green',ls='--',marker='s',ms=6) #square box
plt.plot(Salary[1],color='blue',ls=':',marker='o',ms=6) #square box
plt.plot(Salary[3],color='black',ls='--',marker='^',ms=6) #square box
plt.xticks(list(range(0,10)),Seasons,rotation='vertical') #X axis values
#rotation values as 'vertical'
plt.show()
```



[27]:
```
plt.plot(Salary[0],color='green',ls='--',marker='s',ms=6,label=Players[0])␣
↪#square box
plt.plot(Salary[1],color='blue',ls=':',marker='o',ms=6,label=Players[1])␣
↪#square box
plt.plot(Salary[2],color='purple',ls='--',marker='^',ms=6,label=Players[2])␣
↪#square box
plt.plot(Salary[3],color='black',ls='--',marker='d',ms=6,label=Players[3])␣
↪#square box

plt.legend(loc='upper left',bbox_to_anchor=(0,0))

plt.xticks(list(range(0,10)),Seasons,rotation='vertical') #X axis values
#rotation values as 'vertical'
plt.show()
```



[28]:
```
plt.plot(Salary[0],color='green',ls='--',marker='s',ms=6,label=Players[0])␣
↪#square box
plt.plot(Salary[1],color='blue',ls=':',marker='o',ms=6,label=Players[1])␣
↪#square box
plt.plot(Salary[2],color='purple',ls='--',marker='^',ms=6,label=Players[2])␣
↪#square box
plt.plot(Salary[3],color='black',ls='--',marker='d',ms=6,label=Players[3])␣
↪#square box

plt.legend()

plt.xticks(list(range(0,10)),Seasons,rotation='vertical') #X axis values
#rotation values as 'vertical'
plt.show()
```


1e7
3.0
2.5
2.0
1.5
1.0
0.5
Sachin
Rahul
Smith
Sami
2010
2011
2012
2013
2014
2015
2016
2017
2018
2019

```
[ ]:
```